

AMAN SYED

AI Engineer | RAG | LLM Platforms | Agentic AI | AWS

London, UK | amansyed3342@gmail.com | 07774 704102 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Associate Data Scientist who reduced analyst effort by **95%** and scaled GPU enrichment to **2-3M** articles/day, with production experience owning the design, build, and deployment of agentic workflows and RAG systems used daily by research and policy analysts. Architected Penta Brain, a hybrid intelligence search and conversational RAG service combining **pgvector**, full-text search, NER, and RRF fusion with SSE streaming and citation grounding. Shipped Synthetic Stakeholders, a multi-persona AI simulation platform with **18 FastAPI** routes, OpenAI Responses API, and **pgvector**-backed semantic memory. Built **8+ Python**-based NLP/AI self-service tools and **14 MCP-tool** AI workspaces. Hardened releases with 1,245+ tests and automated 3-provider QA. Experienced translating ambiguous stakeholder requirements into scoped, observable, and auditable AI products from prototype to production.

Experience

Associate Data Scientist

Oct 2024 to Present

Penta Group, London

Global stakeholder solutions firm using proprietary AI, research, and advisory expertise to manage reputation and stakeholder challenges.

- **Production RAG:** Architected enterprise RAG pipeline ingesting **2-3M** daily articles, using pgvector HNSW indexing and hybrid retrieval (vector + tsvector + NER) unified via Reciprocal Rank Fusion, powering **15+** LLM narrative workflows.
- **Transcript Intelligence:** Operationalised multi-agent RevAI/LLM extraction workflows, reducing analyst review from **20** hours to under 5 minutes at **92%** precision for transcript-to-insight generation.
- **Agentic Orchestration:** Engineered multi-agent extraction workflows with document-grounded prompting and structured outputs, surfacing reputational signals across transcript intelligence and stakeholder analysis workflows for automated analyst briefings.
- **Narrative Generation:** Launched **15** narrative endpoints across **15** modules with embeddings, clustering, MMR evidence, and provider fallback for analyst-facing AI delivery.
- **GPU Inference:** Productionised Docker-containerised embedding and NER inference (PyTorch, bfloat16, CUDA) on AWS ECS/EC2 ASGs with SQS-driven autoscaling, processing **2-3M** daily articles across GPU workers.
- **Context Engineering:** Designed K-means/HDBSCAN/MMR clustering pipeline distilling **10,000+** articles into LLM-ready context under token limits, improving answer relevance and cutting prompt context by **62%** across **15+** narrative generation endpoints.
- **Observability Stack:** Established LLM quality monitoring tracking BLEU, ROUGE, BERTScore, and per-class F1/MAE against **12,791-row** gold datasets, exposed via FastAPI with CloudWatch dashboards and request-ID telemetry.
- **Persona Engine:** Built enterprise AI persona simulation platform with OpenAI/Anthropic APIs, RAG, and SSE streaming, exposing **18** REST routes with auth, rate limiting, and pgvector-backed semantic memory for multi-persona conversations.
- **Release Pipelines:** Deployed containerised microservices across int/pilot/prod via AWS CodePipeline, CodeBuild, and ECR, with automated Ruff, mypy, pytest and 3-provider AI release QA gates, enabling zero-downtime ECS rolling updates.
- **Self-Service AI:** Operationalised **8+** Python-based NLP/AI tools on EC2, covering sentiment distribution, influencer identification, storyline detection, and topic analysis for analyst workflows.
- **Sentiment Evaluation:** Improved LLM sentiment classification to **71%** weighted F1 on 1,000 labelled articles by benchmarking **33+** prompt and model configurations, establishing a production NLP classification benchmark.
- **AI Workspaces:** Built AI workspace platform with **14** MCP tools enabling self-serve analysis, automated visualisations, and evidence-backed reporting for insights teams.

Financial Research Assistant

Sep 2024 to Oct 2024

The University of Edinburgh Business School, Edinburgh

Research project evaluating environmental disclosure across large U.S. companies using Refinitiv Workspace data.

- **API Data Pipeline:** Engineered a Python pipeline using the Refinitiv API to automate extraction of 3 years of financial metrics and earnings call transcripts across Fortune **500** U.S. companies, replacing manual Workspace downloads.
- **Text Analytics Preparation:** Structured thousands of corporate disclosures and call transcripts into analysis-ready datasets for NLP-based sentiment and environmental-disclosure scoring.

Data Engineer

Dec 2022 to Aug 2023

C.H. Robinson, Mumbai

Global logistics and supply-chain technology company serving high-volume freight and reporting workflows.

- **Serverless Pipeline:** Designed and implemented a data pipeline using Lambda, S3, and API Gateway to process **100K** daily API records and improve data availability for operational reporting.

- **Workflow Automation:** Automated processes using Selenium and Python scripts, saving **90+** monthly engineering hours and doubling data capture capacity for reporting operations.
- **Logistics Reporting:** Automated daily client report generation with Python, increasing output to **20+** reports daily for improved client logistics visibility.
- **Lambda Optimisation:** Optimised 200+ AWS Lambda functions through SQL query tuning and Python refactoring, improving data processing efficiency by **40%** for production workflows.

Education

MSc Statistics with Data Science	<i>Sep 2023 to Aug 2024</i>
The University of Edinburgh, Edinburgh	<i>Completed</i>
Dissertation: Lloyds Bank Insider Risk Detection - time-series anomaly detection using Isolation Forest, LOF and Autoencoders on imbalanced transactional data; achieved 60% F1 score validated against precision-recall and temporal off-hours risk patterns.	
BEng Computer Engineering	<i>Aug 2019 to Jun 2023</i>
University of Mumbai, Mumbai	<i>9.83/10 CGPA</i>

Technical Skills

- **Agentic AI and LLMs:** RAG, hybrid retrieval, agent orchestration, multi-agent workflows, prompt engineering, LLM routing, multi-provider fallback (OpenAI GPT, Google Gemini, Anthropic Claude), LLM evaluation (BLEU, ROUGE, BERTScore, F1/MAE), embeddings, vector search, semantic caching, MCP, LangChain, Google ADK
- **Backend and APIs:** Python, FastAPI, Flask, Pydantic, SQLAlchemy, REST APIs, SSE streaming, PostgreSQL, pgvector, MySQL, MongoDB, Firebase
- **NLP and ML:** PyTorch, HuggingFace, GLiNER, spaCy, BERT, SentenceTransformers, UMAP, HDBSCAN, LDA/NMF, scikit-learn
- **Cloud and MLOps:** AWS (ECS, SQS, SageMaker, Lambda, S3, SSM, CloudWatch, ECR, CodeBuild, CodePipeline), Docker, Terraform, GitHub Actions, CI/CD
- **Frontend:** React 19, TypeScript, Vite, Tailwind CSS, Recharts
- **Languages:** Python, SQL, TypeScript, JavaScript, R, Java

Projects

Legal Triage Copilot <i>Contract Triage Prototype</i>
<i>Python, FastAPI, Pydantic, OpenRouter, HTML/CSS/JS, evaluation framework</i>
• Triage Engine: Built a dual-mode regex and LLM contract triage system routing NDAs and order forms into 3 risk tiers with clause-by-clause analysis in under 100ms, eliminating manual legal pre-screening.
• Eval Coverage: Designed a 12 -sample evaluation suite with CSV ground truth and JSON reporting, validating pipeline accuracy across all three routing tiers with per-file pass/fail detail.

Finance Flow (frontend backend) <i>Personal Finance Platform</i>
<i>React 19, TypeScript, Vite 6, Tailwind CSS v4, FastAPI, Recharts, Google Sheets API, Frankfurter API, pytest, Render</i>
• Bank Normalisation: Engineered a Strategy Pattern FastAPI backend with full pytest coverage, normalising 6 UK bank CSV formats into a unified 7-field schema, enabling personal finance tracking across all major UK banks from one dashboard.
• Live FX Proxy: Built a secure Frankfurter exchange-rate proxy with caching prevention and timeout safety, plus full Google Sheets CRUD orchestration enabling real-time multi-currency sync.

News RAG <i>Multi-Provider News RAG Pipeline</i>
<i>Python, FastAPI, PostgreSQL, pgvector, Docker Compose, EmbeddingGemma, GLiNER, Bedrock/Vertex Claude, React, MCP</i>
• RAG Pipeline: Built a multi-provider news RAG system with FastAPI, PostgreSQL/pgvector HNSW, EmbeddingGemma embeddings, and GLiNER NER, serving hybrid retrieval (vector + full-text + entity) via RRF fusion with SSE streaming and citation grounding.
• Chat & MCP: Integrated Bedrock/Vertex Claude chat with conversation memory, web search augmentation, and a FastMCP wrapper exposing 10 tools, plus a React UI for interactive querying and conversation history.

Certifications

- AI and Digital Transformation in Government - University of Oxford Saïd Business School and UNESCO
- Architecting with Google Compute Engine Specialization - Google Cloud
- Machine Learning Foundations: A Case Study Approach - University of Washington